

**OBJECT-ORIENTED PROGRAMMING WITH APPLICATIONS TO NUMERICAL METHODS AND
LINEAR ALGEBRA, ES 4075 - 01
SPRING 2026, ASSIGNMENT – 3, # OF PROBLEMS: 3, TOTAL POINTS: 20
SUBMISSION DATE: 03/04/2026 BY 4:00 PM**

NOTE:

- YOU CANNOT TYPE THE CODE IN THIS ASSIGNMENT ON A COMPUTER.
- YOU NEED TO ANSWER BY INSPECTION ONLY.
- **YOUR ANSWERS HAVE TO BE TYPED. HANDWRITTEN SUBMISSIONS WILL NOT BE GRADED.**
- **SUBMIT A SINGLE PDF CONTAINING YOUR ANSWERS.**

Prob. 1

POINTS: 8

In the code below, there are two compiler errors and two run-time errors. Identify the errors, explain the reasons for them, and correct the code.

```
#include <iostream>
#include <iomanip>

int main()
{
    int ptr = 9, * temp, * temp1, value = 5;
    float* My_temp, res = 10.0;
    temp = &ptr;
    temp1 = &value;
    *temp1 = temp + temp1;
    std::cout << "The address of the variable values is" << ptr;
    &My_temp = *res;
    std::cout << "The data stored in the variable res is" << My_temp;
    return 0;
}
```

Prob. 2

POINTS: 4

In the code below, write the output of the cout statements and provide justifications.

```
#include <iostream>

int main()
{
    int count, sum;
    int count1, check;
    count = 2;
    sum = 0;
    count1 = 0;
    while (count < 100)
    {
        check = 8 / count;
        if (check == 10 || check == 5)
```

```

        {
            count1 += 10;
            sum += count1;
        }
        if (check == 2)
        {
            count = 200;
        }
        else {
            count += 2;
        }
    }
    std::cout << "The contents of the variable count is " << count << std::endl;
    std::cout << "The contents of the variable sum is " << sum << std::endl;
    return 0;
}

```

Prob. 3

POINTS: 8

In the code below, there are two compiler errors and two runtime errors on the lines accompanied by comments. Identify the errors and correct the code.

```

#include <iostream>
int main()
{
    int nData1 = 9, nData2 = 5;
    int* pData1, * pData2;
    pData1 = &nData1;
    pData2 = &nData2;
    *pData2 = pData1 + pData2; // Perform addition of nData1 and nData2 and store
the result in nData2.
    std::cout << "The address of the variable nData1 is " << nData1 << endl; //
Display address of nData1
    float fData = 10.0;
    float* pfData;
    &pfData = *fData; // Store the address of fData in the pointer pfData.
    std::cout << "The data stored in the variable fData " << pfData << endl; //
Display the data stored in variable fData using the pointer pfData.
    return 0;
}

```